

```

// separate upper and lower digits
j=offtime%10; // show value 00
// if ontime-offtime flag is not 1
{
    else if (ontime-offtime flag is not 1)
    {
        if (ontime<60) on-
        time++; // increase ontime till 60
        else
        {
            writecmd(0xC0); // same as above
            writestr("60-max limit ");
            t1=l;
            i=ontime/10;
            j=ontime%10;
            if ((t1==0)&& (t2==0)) //
            if max limit is not reached
            {
                writecmd(0xC0); // print ontime/off-
                time sec
                writestr("press enter:");
                writedat(ascii[i]); // as one by one digit
                writedat(ascii[j]);
                writestr("
");
            }
            void dectime()
            {
                // similar to above function
                t1=0;
                if (k==1)
                {
                    if (offtime>1) off-
                    time--;
                    if (offtime<1)
                    {
                        writecmd(0xC0);
                        writestr("01-min limit ");
                        t2=l;
                        i=offtime/10;
                        j=offtime%10;
                        else
                        {
                            if (ontime>1)
                            {
                                if (ontime<1)
                                {
                                    writecmd(0xC0);
                                    writestr("01-min limit ");
                                    t2=l;
                                    i=ontime/10;
                                    j=ontime%10;
                                    if ((t1==0)&& (t2==0))
                                    {
                                        writecmd(0xC0);
                                        writestr("press enter:");
                                        writedat(ascii[i]);
                                        writedat(ascii[j]);
                                        writestr("
");
                                    }
                                    void enter()
                                    {
                                        // enter ontime & offtime and
                                        // display message
                                        k++;
                                        // increase count to enter off time
                                        if (k==1) //
                                        first time
                                        {
                                            writecmd(0x80);
                                            writestr("Set OFF time
                                            and"); // display message and
                                            writecmd(0xC0);
                                            writestr("press en-
                                            ter:");
                                            writedat(0x30);
                                            writedat(0x30);
                                            i=j=0;
                                            else if (k==2)
                                            // second time
                                            {
                                                writecmd(0x01);
                                                // clear LCD screen and
                                                writestr("P Press
                                                start"); // display message
                                                sflag=l;
                                                // set start flag
                                                k=0;
                                                // reset counter
                                                // generate delay of exact 1 sec
                                                int x;
                                                T1 = 0x8F;
                                                T1 = 0x3C;
                                                TRI = 1;
                                                for (x=0;x<=20;x++)
                                                {
                                                    while (T1!=0);
                                                    T1 = 0;
                                                    T1 = 0x8F;
                                                    T1 = 0x3C;
                                                    TRI = 0;
                                                }
                                                void start()
                                                {
                                                    int a,b;
                                                    tmp1=ontime;
                                                    // save ontime and offtime values
                                                    tmp2=offtime;
                                                    back=writecmd(0x80);
                                                    writestr("Device ON: ");
                                                    // display message
                                                    // switch ON relay
                                                    for (a=tmp1;a>0;a--)
                                                    // start countdown again
                                                    {
                                                        i=a/10;
                                                        j=a%10;
                                                        writecmd(0x8B);
                                                        writedat(ascii[i]); // change the display
                                                        digits
                                                        writedat(ascii[j]); // on LCD after
                                                        delay
                                                        lsec(); // every sec
                                                        op=0;
                                                        // switch OFF relay
                                                        writecmd(0x80);
                                                        writestr("Device OFF:");
                                                        for (b=tmp2;b>0;b--)
                                                        // start countdown again
                                                        {
                                                            i=b/10;
                                                            j=b%10;
                                                            writecmd(0x8B);
                                                            writedat(ascii[i]); // display digits after
                                                            delay
                                                            lsec(); // 1 sec delay
                                                            if (pin==1) goto back;
                                                            // repeat continuously for repeat mode
                                                            else
                                                            // otherwise
                                                            {
                                                                flag=l;
                                                                // set flag
                                                                ontime=0;
                                                                // clear ontime and offtime
                                                                offtime=0;
                                                                // and stop operation
                                                            }
                                                        }
                                                    }
                                                    void takeinput() interrupt 2 // for remote
                                                    operation mode
                                                    {
                                                        byte=Pl;
                                                        // take input from Pl when interrupt ar-
                                                        rives
                                                        main()
                                                        {
                                                            TMOD=0x10;
                                                            P3=0x68;
                                                            // set keypad operation and re-
                                                            peat mode
                                                            P2=0x00;
                                                            // P2 as output
                                                            writecmd(0x3C);
                                                            // Initialize LCD
                                                            writecmd(0x0E);
                                                            bgin: writecmd(0x01);
                                                            // clear display
                                                            writestr("Set ON time and");
                                                            // display message and
                                                            writecmd(0xC0);
                                                            writestr("press Enter:");
                                                            writedat(0x30);
                                                            // value as 00
                                                            writedat(0x30);
                                                            IT1=l;
                                                            // enable interrupt on falling
                                                            edge
                                                            IE=0x84;
                                                            // enable ext interrupt 1
                                                            flag=0;
                                                            // clear flag for entering values
                                                            sflag=0;
                                                            // start flag cleared
                                                            agin=Pl=0xFF;
                                                            // Pl as input
                                                            byte=0xFF;
                                                            // initialize byte variable
                                                            if (rmt==1) while (Pl==0xFF); //
                                                            for keypad mode check status of Pl
                                                            else while (byte==0xFF);
                                                            // for remote mode check byte content
                                                            if (rmt==1) byte=Pl;
                                                            // if keypad mode save input in byte
                                                            switch (byte)
                                                            {
                                                                case 0xFE:
                                                                    // for 1st key
                                                                case 0xFD:
                                                                    // for 2nd key in-
                                                                    crease time
                                                                case 0xFB:
                                                                    // for 3rd key de-
                                                                    crease time
                                                                case 0xF7:
                                                                    // for 4th key enter
                                                                    values
                                                            }
                                                            break;
                                                            keydiy();
                                                            // key debounce delay
                                                            if (flag==0) goto agin;
                                                            // till start button is not
                                                            pressed change and
                                                            else goto bgin;
                                                            // enter values otherwise start operation
                                                            again
                                                        }
                                                    }
                                                }
                                            }
                                        }
                                    }
                                }
                            }
                        }
                    }
                }
            }
        }
    }
}

```